



Daffodil International University

Department of Software Engineering

Faculty of Science & Information Technology

Midterm Examination, Fall 2024

Course Code: ENG-101; Course Title: English-I

Sections & Teachers: All Sections; MA, SST, SSS, AJS

Time: 1 Hour 30 Mins

Marks: 25

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

Read the Passage Carefully and Answer the Questions that Follow:

A Brief (and Tasty) History of Chocolate

Most people today think of chocolate as something sweet to eat or drink that can be easily found in stores around the world. It might surprise you that chocolate was once highly treasured. The tasty secret of the cacao (Kah Kow) tree was discovered 2,000 years ago in the tropical rainforests of the Americas. The story of how chocolate grew from a local Mesoamerican beverage into a global sweet encompasses many cultures and continents.

Historians believe the Maya people of Central America first learned to farm cacao plants around two thousand years ago. The Maya took cacao trees from the rainforests and grew them in their gardens. They cooked cacao seeds, then crushed them into a soft paste. They mixed the paste with water and flavorful spices to make an unsweetened chocolate drink. The Maya poured the chocolate drink back and forth between two containers so that the liquid would have a layer of bubbles or foam.

Cacao and chocolate were an important part of Maya culture. There are often images of cacao plants on Maya buildings and art objects. Ruling families drank chocolate at special ceremonies. And, even poorer members of society could enjoy the drink once in a while. Historians believe that cacao seeds were also used in marriage ceremonies as a sign of the union between a husband and a wife.

The Aztec culture in current-day Mexico also prized chocolate. But cacao plants could not grow in the area where the Aztecs lived. So, they traded to get cacao. They even used cacao seeds as a form of money to pay taxes. Chocolate also played a special role in both Maya and Aztec royal and religious events. Priests presented cacao seeds and offerings to the gods and served chocolate drinks during sacred ceremonies. Only the very wealthy in Aztec societies could afford to drink chocolate because cacao was so valuable. The Aztec ruler Montezuma was believed to drink fifty cups of chocolate every day. Some experts believe the word for chocolate came from the Aztec word "xocolatl" which in the Nahuatl language means "bitter water." Others believe the word "chocolate" was created by combining Mayan and Nahuatl words.

The explorer Christopher Columbus brought cacao seeds to Spain after his trip to Central America in 1502. But it was the Spanish explorer Hernando Cortes who understood that chocolate could be a valuable investment. In 1519, Cortes arrived in current-day Mexico. He believed the chocolate drink would become popular with Spaniards. After the Spanish soldiers defeated the Aztec empire, they were able to seize the supplies of cacao and send them home. Spain later began planting cacao in its colonies in the Americas in order to satisfy the large demand for chocolate. The wealthy people of Spain first enjoyed a sweetened version of chocolate drink. Later, the popularity of the drink spread throughout Europe. The English, the Dutch and the French began to plant cacao trees in their own colonies. Chocolate remained a drink that only wealthy

people could afford to drink until the eighteenth century. Later during the period known as the Industrial Revolution, new technologies helped make chocolate less costly to produce.

Farmers grow cacao trees in many countries in Africa, Central and South America. The trees grow in the shady areas of the rainforests near the Earth's equator. But these trees can be difficult to grow. They require an exact amount of water, warmth, soil and protection. After about five years, cacao trees start producing large fruits called pods, which grow near the trunk of the tree. The seeds inside the pods are harvested to make chocolate. There are several kinds of cacao trees. Most of the world's chocolate is made from the seed of the forastero tree. But farmers can also grow criollo or trinitario cacao plants. Cacao trees grown on farms are much more easily threatened by diseases and insects than wild trees. Growing cacao is very hard work for farmers. They sell their harvest on a futures market. This means that economic conditions beyond their control can affect the amount of money they will earn. Today, chocolate industry officials, activists, and scientists are working with farmers. They are trying to make sure that cacao can be grown in a way that is fair to the timers and safe for the environment.

To become chocolate, cacao seeds go through a long production process in a factory. Workers must sort, clean and cook the seeds. Then they break off the covering of the seeds so that only the inside fruit, or nibs, remain. Workers crush the nibs into a soft substance called chocolate liquor. This gets separated into cocoa solids and fat called cocoa butter. Chocolate makers have their own special recipes in which they combine chocolate liquor with exact amounts of sugar, milk and cocoa fat. They finely crush this "crumb" mixture in order to make it smooth. The mixture then goes through two more processes before it is shaped into a mold form.

Chocolate making is big business. The market value of the yearly cacao crop around the world is more than five billion dollars. Chocolate is especially popular in Europe and the United States. For example, in 2005, the United States bought 1.4 billion dollars' worth of cocoa products. Each year, Americans eat an average of more than five kilograms of chocolate per person. Specialty shops that sell costly chocolates are also very popular. Many offer chocolate lovers to take the chance to taste chocolates grown in different areas of the world.

1.	a)	Classify the following statements as True/False/Not Given based on your reading of the passage. i. Use of cacao and chocolate in ceremonies were restricted in Maya royal families. ii. The Spanish explorer Hernando Cortes invested in chocolate and chocolate drinks. iii. Some parts in cacao seed are get rid of during the process of making chocolate. iv. Chocolate is popular more in some countries or continents than other parts around the world.	[4x1=4]	CLO- 3 Level- C4
	b)	Complete the flowchart below with information from the passage. Don't write MORE THAN TWO WORDS for each answer. <pre> Cacao seeds ↓ sorting, (i). _____ and cooking to get rid of the (ii). _____ ↓ Nibs ↓ Crushing </pre>	[6x0.5=3]	

		↓ (iii). _____ ↓ Add sugar, milk and (iv). _____ ↓ (v). _____ mixture ↓ Crush finely then come into a shape in a (vi). _____ ↓ Chocolate		
2.	a)	Identify an appropriate article (a/an/the/put cross for no article) to fill in the gaps. Last summer, I traveled to (i) _____ Europe for the first time. I started my journey in (ii) _____ Switzerland. I stayed in (iii) _____ charming village nestled in (iv) _____ Alps. Every morning, I enjoyed a fresh coffee and a croissant from (v) _____ local bakery. One day, I decided to explore (vi) _____ nearby lake that was known for its crystal-clear water. I rented (vii) _____ small boat and spent the afternoon paddling around, admiring (viii) _____ breathtaking scenery. By the end of my trip, I felt refreshed and inspired by (ix) _____ serene environment. It was (x) _____ unforgettable experience.	[10x0.5=5]	CLO- 1 Level- C1
	b)	Match the subjects of the following sentences with their correct form of verbs in the brackets. i. He _____ (complete) his homework before his friends arrived. ii. The moon _____ (cause) the ocean tides. iii. Fifty kilometers _____ (be) a long distance to walk. iv. The chairperson and the committee members usually _____ (decide) the monthly plans of the company. v. The Philippines _____ (be) a country of islands.	[5x1=5]	
3.		Construct a paragraph on the following topic Your Neighborhood Or Your First Day at University	[1x8=8]	CLO- 4 Level- C3



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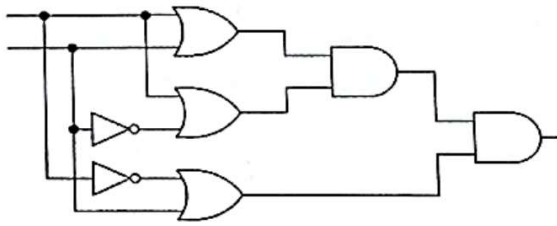
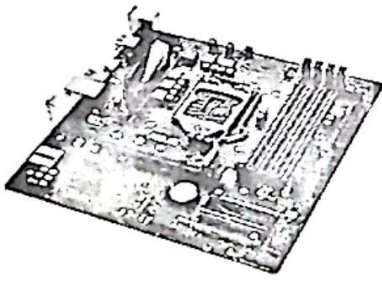
Course Code: SE 111; Course Title: Computer Fundamentals
Sections & Teachers: ALL

Time: 1 Hour 30 Mins

Marks: 25

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	a)	A component of a computer system processes information just like how our brain controls our body, performing various data processing operations. Describe the component mentioned here.	3	CLO-1 C2
	b)	Imagine a computer memory that temporarily stores data only while powered on. Once the power is off, all the data vanishes. But there's another type of storage that retains data even when the power is off. Explain the key differences between these two types of memory in terms of their functionality and data retention.	3	
	c)	I) Convert the 73124.14_8 octal number to its decimal equivalent. II) Convert the 3005_{10} decimal number to its hexadecimal equivalent. III) Convert the $A9B31F_{16}$ hexadecimal number to its octal equivalent.	9	
2.	a)	Illustrate a logic circuit for the Boolean expression F. $F = (X'+Y'+Z) + (X+Y'+Z')(X'Y'Z)(X+Y+Z)'$	2	CLO-1 C4
	b)	Identify the Boolean equation for the below circuit, structure the simplified form of the Boolean expression, and prove your simplification using a truth table. 	5	
	c)	Identify the following component and explain its role in the computer system. 	3	



Daffodil International University

Department of Software Engineering
Faculty of Science & Information Technology
Midterm Examination, Fall 2024

Course Code: SE 113; Course Title: Introduction to Software Engineering

Sections & Teachers: All & AP, DAKM, MKS, JIC, TH, SAN, MIM, ST

Time: 1 Hour 30 Mins

Marks: 25

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

- 20 / 25
1.
 - a) Define Software Engineering and detail the core responsibilities of software engineers. [Marks-3] CLO-1 Level-2
 - b) Describe the work scopes of a Software Engineer and a Software Developer with real-life examples, highlighting their key responsibilities and differences. [Marks-3] CLO-1 Level-2
 - c) Define functional requirements and non-functional requirements of software. Suppose, Daffodil International University is developing a Bus Seat Booking System where students can pre-book seats, view real-time availability, and get booking confirmations. The system aims to reduce overcrowding and improve bus service efficiency. Classify the key functional and non-functional requirements for this system. [Marks-3] CLO-1 Level-2
 - d) Distinguish between High-Level Design and Low-Level Design. [Marks-2] CLO-1 Level-2
 - e) Contrast Quality Assurance and Quality Control with an example. [Marks-3] CLO-1 Level-2
 - f) You are leading a project to develop a comprehensive e-learning platform for a university. The project involves various stakeholders, including educators, students, and administrators, each with specific needs and expectations. Demonstrate the different phases of Software Project Management you would implement to guide the project from start to finish. For each phase, including the different activities you would perform, how you would manage changing requirements, and the tools you would use to ensure the project is completed on time, within budget, and meets the quality standards required by the university. [Marks-4] CLO-2 Level-3
 2.
 - a) Imagine you are leading a team to develop a new banking system with features like secure transactions, user interfaces, and risk management. Given the need for ongoing risk analysis, prototype development, and continuous feedback. Detect a suitable Software Process model considering these points. Describe the activities of each phase for the model. [Marks-5] CLO-2 Level-3
 - b) Imagine you are managing a software application for inventory management in a retail company. After deployment,
 - i. Users report a bug that prevents the system from updating inventory levels accurately.
 - ii. The company's IT infrastructure has been upgraded to a new operating system, and the software needs to be compatible with this new environment.
 - iii. Users also request new features, such as advanced reporting and data analytics, to improve their experience.Determine which type of maintenance to apply for above mentioned scenarios. [Marks-2] CLO-2 Level-3



Daffodil International University

Department of Software Engineering
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Midterm Examination, Fall 2024

Course Code: MAT 101; Course Title: Mathematics I

Sections & Teachers: A, B, C, D, E, F (SB), G, H, I, J, K, L (SKI), M, N (NA), O, P, Q,
Retake (GRS),

Time: 1 Hour 30 Mins

Marks: 25

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	a)	Explain the followings with example: i) Domain and range of function. ii) Periodic function. iii) Even function. iv) One-one function.	3/	[04]	CLO-1 Level-C2
	b)	$f(x) = \begin{cases} 2x + 3; & x < 0 \\ x^2; & 0 \leq x \leq 2 \\ \sqrt{x-2} & x > 2 \end{cases}$ is a piecewise function. i) Identify the domain and range of the given function. ii) Graphically show the function.	5/	[05]	
2.	a)	A function is given by $f(x) = x $. Examine the continuity and differentiability of $f(x)$ at $x = 0$.	u/	[06]	CLO-2 Level-C3
3.	a)	Differentiate : i) $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$ with respect to $\sin^{-1} \frac{2x}{1+x^2}$ ii) $y = \frac{\cos x - \sin x}{\sqrt{1-\sin 2x}}$ with respect to x .	u/	[04]	CLO-2 Level-C3
	b)	If $y = \cos\{\ln(1+x)\}$, then show that $(1+x)^2 y_{n+2} + (2n+1)(1+x)y_{n+1} + (n^2+1)y_n = 0$.	6	[06]	



Daffodil International University
Department of Software Engineering
Faculty of Science & Information Technology
Midterm Examination, Fall-2024
Course Code: BNS-101, Course Title: Bangladesh Studies
Batch: 43_ALL

Time: 1 Hour 30 minutes

Marks: 25

Answer ALL Questions

[The figures in the right margin indicate the full marks and corresponding course outcomes. All portions of each question must be answered sequentially.]

1.	Interpret the demographic dividend's potential perspectives and challenges for the Bangladesh context of translating living standards and reducing unemployment.	05	CLO-1 Level-2
2.	Demonstrate the reasons behind the arrival of foreigners in Bengal and their contribution to the development trend of Bangladeshi identity.	05	CLO-1 Level-3
3.	Answer the following question briefly: a. Illustrate two primary goals of Bangladesh's foreign policy. b. Correlate the two determinates of Bangladesh Foreign Policy. c. Identify the expended forms of NATO and UNCLOS. d. Explain: What is multilateralism? e. Detect the three challenges of Bangladesh's foreign policy.	5×2 =10	CLO-2 Level-4
4.	Clarify the essentials of a good constitution to ensure the democratic right and level of accountability in Bangladesh's aspirations of <i>The July Revolution-2024</i> .	05	CLO-1 Level 2